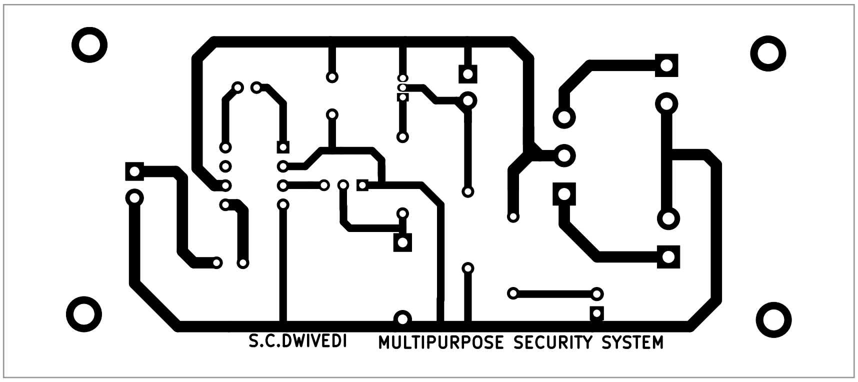

Fig. 3: Actual-size PCB layout

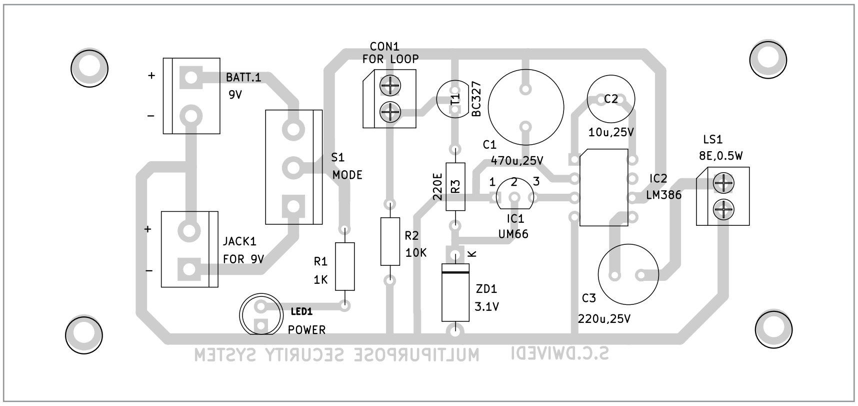

Fig. 4: Components side of the PCB

To amplify the signals, adjust the gain factor through pins 1 and 8. If left open, the gain will be 20dB, as internally set in the IC. By adding a 10µF capacitor (C2) between pins 1 and 8, the gain can be varied up to 200dB. Feed the required signals to pin 3 of IC2.

The musical signal output from IC1 is connected to pin 3 to obtain amplified output on pin 5 of IC2. Capacitor C3 acts as a coupling capacitor, removing DC signals and passing audio signals through it. The amplified signals are then output through speaker LS1.

First, connect a flexible wire as a loop between CON1 and the article (such as luggage) to be protected. The system operates with either a 9V battery or 9V adaptor, depending on the application. After connecting a battery, switch S1 probe towards the battery side. If LED1 glows, the system is ready to use.

After using the power source, keep the mode switch towards the source used. Now, 9V is available in the system. When someone breaks the loop, transistor T1 starts conducting and IC1 receives 3.1V to activate it.

In this system, a flexible wire loop is interfaced as a sensor with the musical IC UM66. LED1 is used in the system as a power supply indicator.

The flexible wire has two terminals which need to be connected across CON1 via the item during security measures. When the loop is cut or broken, it provides a low voltage at the base of transistor T1. Transistor T1 then conducts, enabling IC1 to activate the melody generator sound. The output of IC1 is further directed to IC2 for louder sound.

PARTS LIST

Semiconductors:	
IC1	- UM66 melody generator
IC2	- LM386 low power amplifier
T1	- BC327 PNP transistor
ZD1	- 3.1V zener diode
LED1	- 5mm LED
Resistors (all 1/4-watt, ±5% carbon):	
R1	- 1-kilo-ohm
R2	- 10-kilo-ohm
R3	- 220-ohm
Capacitors:	
C1	- 470µF, 25V electrolytic
C2	- 10µF, 25V electrolytic
C3	- 220µF, 25V electrolytic
Miscellaneous:	
JACK1	- DC socket
S1	- SPDT switch
CON1	- 2-pin connector
BATT.1	- 9V battery
LS1	- 8-ohm, 0.5-watt speaker
	- 9V DC adaptor
	- Flexible wire for loop (length as per requirement)

Construction and testing

An actual-size, single-side PCB layout for the musical security alarm is shown in Fig. 3, with its component layout displayed in Fig. 4.

After assembling the circuit on the PCB, enclose it in a suitable box. Fix LED1 and CON1 on the front side of the box, and the mains power connector JACK1 and 9V battery on the rear side. Connect the flexible wire loop between points A and B (refer to CON1 in Fig. 1) via the item to be secured, ensuring that the loop would break easily when someone attempts to tamper with it.

Bonus. You can watch the video of the tutorial of this DIY project at *https://www.electronicsforu.com/videos-slideshows/make-multipurpose-security-system* EFY

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